LIMITED-DISCLOSURE RESEARCH CONCEPT

INVARIANT-SC

Learning Shared Supply-Chain Meaning from Physical Consistency

A concise faculty brief for AI-enabled sustainability and resilience

CONCEPT IN ONE SENTENCE

INVARIANT-SC studies whether AI can reconcile incompatible supply-chain records by using the physical consistency of material and inventory flows as guidance, creating a common and auditable foundation for sustainability and resilience analysis.

Document purpose: explain the research direction without disclosing implementation-sensitive details.
Disclosure level: problem, intuition, research contribution, evaluation logic, and expected impact only.

1. Executive Overview

Supply chains generate data across procurement, inventory, production, transport, returns, and sustainability systems. Even when organizations describe the same physical event, their databases may use different names, structures, units, identifiers, and timing conventions. This makes cross-company analysis expensive and fragile. It also limits the quality of carbon accounting, waste measurement, inventory visibility, and disruption analysis.

INVARIANT-SC proposes a different source of guidance: the real process beneath the databases must remain physically coherent. Goods enter, move, transform, accumulate, and leave. Inventory changes should be consistent with those events. Instead of trusting labels alone, the proposed research asks an AI system to use these shared physical regularities to infer a common interpretation of heterogeneous records.

CENTRAL RESEARCH QUESTION

Can AI recover a reliable common representation of supply-chain activity when the available data uses different or incomplete semantics, while also recognizing situations where the evidence is insufficient?

Why this matters

- **Data integration:** reduce dependence on manually maintained mappings across organizations and systems.
- **Sustainability:** improve the consistency and traceability of activity data used for emissions and waste analysis.
- **Resilience:** reconstruct a common stock-and-flow view needed to study disruptions, shortages, and recovery.
- **Trust:** retain evidence and uncertainty instead of silently forcing a single interpretation.

Scope of the research

The research focuses on learning a reusable semantic foundation. It is not presented as a full enterprise software platform, a universal carbon calculator, or an autonomous procurement agent.

2. The Core Idea

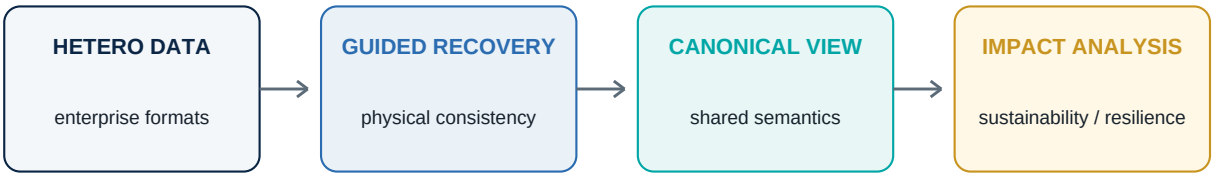

Figure 1. High-level concept flow; implementation details intentionally omitted.

2.1 From local labels to global consistency

Conventional matching methods usually compare names, descriptions, or value patterns. INVARIANT-SC adds a global question: does the proposed interpretation describe a physically plausible supply-chain process over time? A mapping that looks reasonable in isolation should be rejected if it creates persistent contradictions across inventory, movement, transformation, or return records.

2.2 A common flow representation

The intended output is a canonical view of the underlying process that can connect equivalent activities across different data sources. The system should preserve data lineage, identify uncertain mappings, and support human review where multiple interpretations remain plausible.

2.3 Responsible uncertainty

A key research principle is that the model must be allowed to say that the available observations do not uniquely determine the answer. This avoids producing a precise-looking but unsupported data integration.

HIGH-LEVEL HYPOTHESIS

Physical consistency can provide a transferable supervisory signal that remains useful even when headers, schemas, or enterprise conventions change.

What this brief intentionally withholds

- The internal program representation and optimization objective.
- The exact search, learning, and verification mechanism.
- Formal theorem assumptions, proof strategy, and ambiguity construction.
- Benchmark-generation recipes, stress parameters, and private experimental results.
- Engineering architecture, thresholds, and implementation roadmap.

3. Research Contribution and Novelty

Schema matching, semantic column typing, conservation-aware learning, traceability standards, and supply-chain modeling already exist as mature research areas. The proposed novelty is their new combination around a single learning problem: recovering a globally coherent physical-process interpretation from heterogeneous relational observations, with explicit handling of unresolved meaning.

Existing direction	Typical focus	Proposed distinction
Schema matching	Find corresponding fields across data sources.	Require the complete interpretation to agree with the physical process over time.
Semantic typing	Assign a known label to an individual column.	Recover a connected process representation rather than independent labels.
Relational learning	Learn from tables whose entities and links are already known.	Address the upstream problem of establishing a trustworthy common meaning.
Conservation-aware ML	Improve prediction when the physical model is already specified.	Use physical consistency to help infer the data interpretation itself.
Traceability standards	Define how interoperable event data should be represented.	Help reconcile legacy records toward a shared, auditable representation.

Proposed contribution types

1. A new benchmarkable task for invariant-guided semantic recovery under schema variation.
2. A learning framework that combines data evidence with global process consistency.
3. A principled treatment of cases where more than one interpretation fits the observations.
4. A cross-schema evaluation setting designed to test generalization rather than memorization.
5. Downstream validation in sustainability and resilience applications.

NOVELTY POSITION

The defensible claim is not 'AI for supply chains' in general. It is the use of shared flow consistency to recover and audit latent data meaning across incompatible supply-chain records.

4. Evaluation and Expected Impact

4.1 Evaluation logic

The research will be evaluated on whether the recovered common representation remains correct when labels, database layouts, organizations, process structures, data quality, and timing conventions change. The study will compare against strong language-model, schema-matching, semantic-typing, join-discovery, heuristic, and optimization baselines.

- Recovery accuracy for semantic roles and relationships.
- Generalization to previously unseen schemas and process structures.
- Robustness to missing, delayed, or noisy records.
- Quality of uncertainty and abstention when evidence is ambiguous.
- Value of the reconstructed representation for downstream analyses.

4.2 Sustainability applications

The canonical flow view can support more consistent activity data for material, transport, energy, returns, and waste. Emission factors and reporting boundaries remain externally governed; the research contribution is to improve the traceability and semantic consistency of the underlying activity records.

4.3 Resilience applications

A common view of inventory and movements can improve visibility into shortages, disruption propagation, recovery, and multi-tier dependencies. These applications test usefulness without becoming separate headline contributions.

4.4 Expected research value

Research value	Expected outcome
Scientific	A general problem at the intersection of representation learning, program induction, relational data, and physical supervision.
Operational	Lower-friction reconciliation of heterogeneous supply-chain records with visible evidence and uncertainty.
Sustainability	More auditable activity data for carbon, waste, and circular-flow analysis.
Resilience	A consistent data foundation for inventory visibility and disruption studies.

RECOMMENDATION

Proceed with INVARIANT-SC as a focused research direction. Keep the paper centered on invariant-guided semantic recovery and uncertainty; use sustainability and resilience as coherent evidence of impact.

5. Selected Public References

The following public sources establish the surrounding research landscape. They do not disclose the proposed implementation.

[1] N. Seedat and M. van der Schaar. Bootstrapping Self-Improvement of Language Model Programs for Zero-Shot Schema Matching. ICML, 2025. PMLR page.

[2] M. Fey et al. Position: Relational Deep Learning - Graph Representation Learning on Relational Databases. ICML, 2024. PMLR page.

[3] S. Chang et al. Learning Production Functions for Supply Chains with Graph Neural Networks. AAAI, 2025. AAAI page.

[4] National Institute of Standards and Technology. Enabling and Using Traceability and Data Linking for Sustainable and Efficient Supply Chains. NIST project.

[5] GS1. EPCIS Standard, version 2.0.1. Official standard.

[6] GHG Protocol. Scope 3 Calculation Guidance. Official guidance.

DISCLOSURE NOTE

This document is intentionally limited to the research problem, intuition, contribution boundary, and evaluation philosophy. It omits implementation-sensitive details and unpublished experimental evidence.

Provisional research position: the direction is sufficiently differentiated to justify deeper validation, but novelty and venue suitability remain subject to systematic prior-art review and experimental evidence.